Electromagnetic Waves

Exam code: 0625 & 0972

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Electromagnetic waves

- The electromagnetic spectrum is arranged in a specific order based on the wavelengths or frequencies of the radiation in each region
  - Regions with a **long wavelength** have a **low frequency** and vice versa
- The main regions of the continuous electromagnetic (EM) spectrum from longest to shortest wavelength are:
  - Radio waves
  - Microwaves
  - Infrared
  - Visible (red, orange, yellow, green, blue, indigo, violet)
  - Ultraviolet
  - X-rays
  - Gamma rays

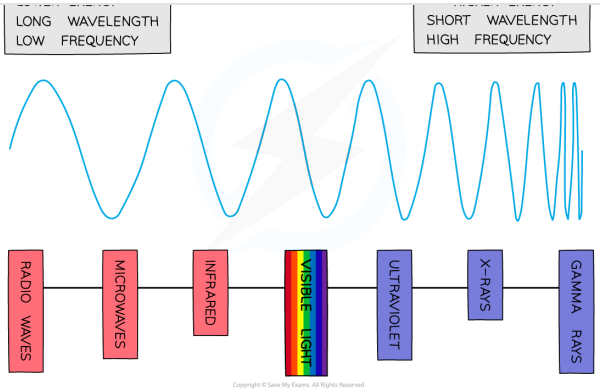
The electromagnetic spectrum in order of decreasing wavelength

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Visible light is just one small part of a much bigger spectrum: The electromagnetic spectrum

Examiner Tips and Tricks

See if you can make up a mnemonic to help you remember the EM spectrum!

One possibility is:

Raging **M**artians **I**nvaded **V**enus **U**sing **X**-ray **G**uns

The electromagnetic spectrum is usually given in order of **decreasing wavelength** and **increasing frequency** i.e. from radio waves to gamma waves

Remember:

- Radios are **big** (long wavelength)
- Gamma rays are emitted from atoms which are **very small** (short wavelength)

Properties of electromagnetic waves

- All electromagnetic waves travel at the same **high speed** in a **vacuum**
- Electromagnetic waves are defined as:

Transverse waves that transfer energy from the source of the waves to an absorber

The Speed of Electromagnetic Waves

Extended tier only

- This is approximately the same speed as electromagnetic waves in air

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